

Minimizing Risk in Mushroom Foraging: Precision-Oriented Classification with Limited Observable Features

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Introduction

Distinguishing between edible and poisonous mushrooms is a classification problem of high practical relevance, as misidentification can have serious consequences for human health [1]. In this context, the application of Data Science and Machine Learning techniques supports the decision-making process through models capable of systematically analyzing the morphological characteristics of mushrooms and assessing their edibility.

This paper analyzes the Mushroom dataset [2] using a multi-level structured approach, combining supervised classification models and unsupervised clustering methods. The objective is to evaluate both predictive performance and the models' ability to operate in realistic scenarios characterized by partial information and limited feature observability.

In the first phase, the problem is approached in a "naïve" manner, addressing the most basic question typically posed in classification problems: which classification model is most effective at distinguishing edible mushrooms from poisonous ones using the entire set of predictors available in the dataset? This analysis provides a benchmark for the maximum achievable performance in an ideal scenario where all variables are considered observable and informative. In parallel, an unsupervised clustering approach is evaluated on the full feature set to determine if the intrinsic data structure allows the dataset to be partitioned into homogeneous, ideally "pure," groups containing exclusively edible

or exclusively poisonous mushrooms, and to identify the minimum number of clusters required to avoid contamination between the two classes.

Subsequently, the study introduces an application-domain-oriented perspective. First, a domain expert identified the features within the dataset that are actually assessable during foraging. These were categorized into: features easily observable even by a novice; features observable with varying degrees of difficulty due to specific environmental conditions; and features not directly evaluable in the field. In this context, the research question becomes: Do easily observable characteristics alone allow for the construction of a classification model capable of correctly distinguishing edible from poisonous mushrooms? To what extent does model performance vary when the variables used are minimized?

Finally, the study explores a pathway simulating the transition from an inexperienced user to a progressively more competent one through an incremental feature addition procedure. Starting from the minimum set of easily assessable characteristics and gradually adding those considered more difficult to observe, a further research question is addressed: How much does classification capacity improve as the complexity of the considered features increases, and to what extent does the progressive addition of new variables allow the model to approach the performance achieved in the ideal scenario?

Throughout all phases of the analysis, model evaluation is oriented toward risk minimization, prioritizing precision as the

evaluation metric. This choice is driven by the need to limit, if not eliminate, the presence of poisonous mushrooms among those classified as edible, even at the cost of reducing the overall coverage of predictions.

Dataset and pre-processing

The dataset utilized for this study is the Mushroom Classification Dataset, made available by the UCI Machine Learning Repository [2] and accessible on the Kaggle platform in CSV format [3]. The dataset contains 8,124 observations, each corresponding to a mushroom specimen described through a set of morphological and structural characteristics belonging to 23 species of gilled mushrooms in the *Agaricus* and *Lepiota* families. Each species is identified as definitely edible or definitely poisonous through the target variable class, which can take only two values: e (edible) or p (poisonous). The original dataset comprises 22 predictive variables in addition to the target variable. All variables are nominal categorical, encoded via single letters, each representing a specific category [table 1, figure 1].

In the initial phase, an exploration of the dataset was conducted to verify its consistency. It was found that one variable used the "?" symbol to indicate missing information. These values were converted to missing values (*Null*) to enable consistent handling in subsequent analysis phases. Consequently, two variables were removed: veil-type, as it presents a single category for all observations and thus lacks informative value, and stalk-root, given that it presents a high number of missing values (> 30%). Furthermore, from an application domain perspective, stalk-root represents a characteristic that cannot be sensibly imputed, as it describes an intrinsic physical property of the mushroom rather

than a probability; moreover, according to the domain expert, it is a characteristic that is not directly assessable in the field. For these reasons, and in alignment with the project's objectives regarding risk reduction and feature observability, the variable was excluded from the analysis. An assessment of the target class distribution reveals a nearly perfect balance between edible (52%) and poisonous (48%) specimens; therefore, no class imbalance issues are present.

Subsequently, the correlation between the remaining categorical variables was evaluated using Cramér's V index [4]. The choice of this index is motivated by the fact that all variables are nominal categorical. The analysis highlighted [figure 2] that the variable gill-attachment presents a very high correlation with stalk-color-above-ring and stalk-color-below-ring (Cramér's $V > 0.97$), as well as with veil-color (Cramér's $V > 0.95$). These values indicate information redundancy; consequently, correlation-based feature selection was performed by removing the gill-attachment attribute from the dataset.

Finally, the association between the target variable class and each of the variables was evaluated, again using Cramér's V index, to analyze which characteristics showed the strongest relationship with mushroom edibility [figure 3]. This exploratory analysis was used to understand the informative contribution of individual variables and to support subsequent modeling phases, without introducing automatic selection based exclusively on correlation with the target variable.

Variable	Number of Categories	Observability
cap-shape	6	e.o.
cap-surface	4	e.o.
cap-color	10	e.o.
bruises	2	e.o.
odor	9	d.o. (5)
gill-attachment	2	d.o. (e)
gill-spacing	2	d.o. (4)
gill-size	2	e.o.
gill-color	12	e.o.
stalk-shape	2	e.o.
stalk-root	7	n.e. (e)
stalk-surface-above-ring	4	d.o. (2)
stalk-surface-below-ring	4	d.o. (1)
stalk-color-above-ring	9	e.o.
stalk-color-below-ring	9	e.o.
veil-type	1	n.e. (e)
veil-color	4	e.o.
ring-number	3	e.o.
ring-type	8	d.o. (3)
spore-print-color	9	n.e.
population	6	e.o.
habitat	7	e.o.

Table 1 - Variables present in the dataset with indication of the number of categories and the classification on observability by the domain expert: e.o. - easily observable, d.o. - difficult to observe, n.e. not evaluable. The numbers in the classification indicate the order of difficulty from 1, easiest, to 5, most difficult. (e) indicates the features excluded during the preprocessing phase.

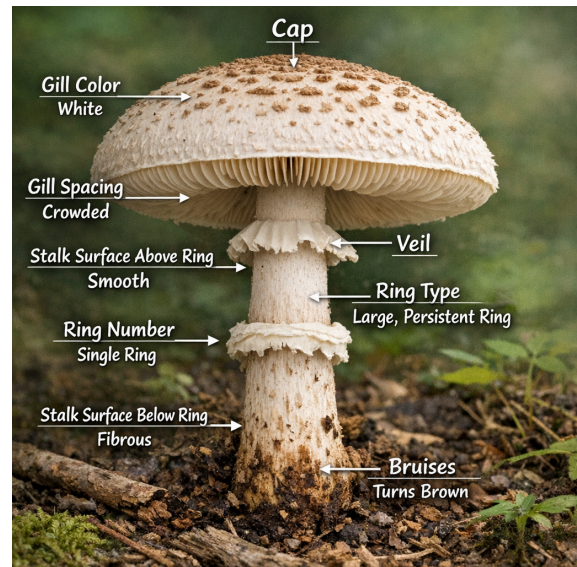


Figure 1 - Example of a mushroom and its physical characteristics (AI-generated image)

Cramér's V Correlation Matrix

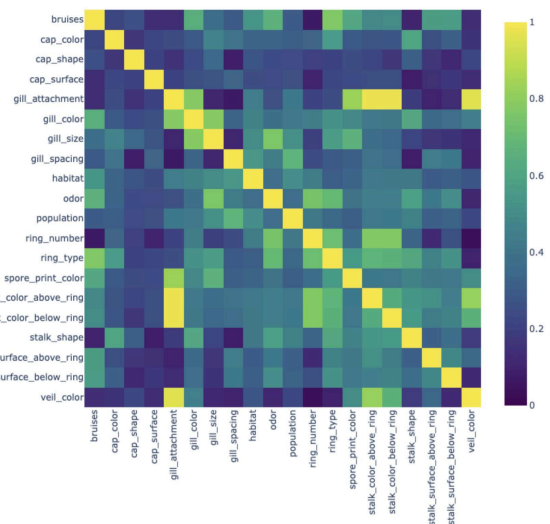


Figure 2 - Correlation matrix among the available variables calculated using Cramér's V. The variables veil-type and stalk-root are not present as they were already excluded.

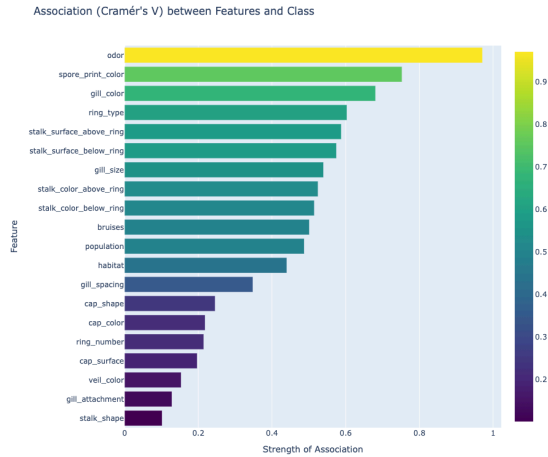


Figure 3 - Association of the variables of the complete cleaned dataset with the target variable through the evaluation of Cramér's V.

Following data cleaning, missing value handling, and the reduction of redundant features, the final dataset consists of 19 categorical variables aligned with the project objectives and suitable for both supervised classification models and unsupervised clustering analyses. The cleaned dataset (hereinafter referred to as “complete”) constitutes the common basis for all subsequent analyses.

Methodology and Analysis

As stated in the introduction, this work aims to evaluate potential support for a novice approach to identifying mushroom edibility. A study has demonstrated that a fixed selection of variables yields better performance than a random choice [5]; for this reason, the various characteristics were divided into three groups by a domain expert: easily identifiable (e.o), difficult to identify (d.o), and not evaluable (n.e.). The former were used as the minimum baseline set, and the non-evaluable ones were excluded. Meanwhile, an order of difficulty was established for the difficult-to-evaluate features (1 - simplest, 5 - most difficult), which were then used to perform feature integration upon the minimum baseline set (see Table 1).

Baseline with all Variables

In the first phase of the analysis, the classification problem of edible and poisonous mushrooms was addressed using the entire set of features available in the complete dataset, in order to establish a reference baseline for the maximum performance achievable in an ideal scenario. Model evaluation was conducted using a 10-fold cross-validation procedure stratified by the target variable, to obtain a robust and stable performance estimate and avoid dependencies on a single data split. Several supervised classification models were trained and compared, representing conceptually different approaches: decision trees (J48), Naive Bayes Tree, Naive Bayes, Support Vector Machine, and logistic regression [6]. Performance evaluation focused on minimizing the risk of predicting a poisonous mushroom as edible, thus prioritizing precision as the primary metric (with “e” as the positive class), as this is suited for limiting the misclassification of poisonous mushrooms as edible.

Minimum Set of Observable Variables

In the second phase, the entire analysis was repeated by limiting the dataset to only those variables identified by the domain expert as easily observable during mushroom foraging, referred to as the base dataset. In this scenario, all supervised classification models considered in the first part were retrained and evaluated on this minimum feature set to analyze the impact of information reduction on predictive performance. Consistent with the previous phase, model evaluation was conducted using 10-fold cross-validation, maintaining precision as the reference metric to ensure direct comparability with the results obtained in the baseline phase. This approach allows for determining whether the loss of information leads to a significant

reduction in the ability to correctly distinguish edible from poisonous mushrooms and to what extent the risk increases compared to the baseline.

Feature increasing

In the third phase, the analysis focused on evaluating the impact of the progressive addition of variables to the base dataset. Starting from only the easily observable characteristics, variables classified by the domain expert as difficult to evaluate were gradually introduced, following an order of increasing complexity and culminating in the complete dataset. The objective of this phase is to analyze how the performance of the supervised models varies by evaluating the variation in precision as the number of features increases.

In parallel, the number of groups necessary to obtain pure sets in hierarchical clustering was observed: at each iteration, variations in entropy with respect to the class variable and in the number of clusters were monitored, systematically comparing the results with those of the baseline defined in the first phase.

Results and Discussion

The analysis of the results obtained with the supervised classification models clearly highlights the high separability between edible and poisonous mushrooms in the Mushroom Classification dataset, especially when the entire set of variables is available.

Complete Dataset

Using the complete dataset, the precision values obtained are extremely high for most of the models considered [Figure 4]: J48, Logistic Regression, Support Vector Machine, and Naive Bayes Tree achieve a precision of 1.0, indicating the absence of poisonous mushrooms among those classified as edible. The Naive Bayes

classifier deviates from this trend with a precision of approximately 0.935, highlighting greater sensitivity to the assumption of conditional independence among variables. These results confirm that, in an ideal scenario where all features are available, the classification problem is substantially solvable without ambiguity, even by relatively simple models such as decision trees. Therefore, this performance represents a theoretical upper bound and serves as the reference baseline for subsequent analyses.

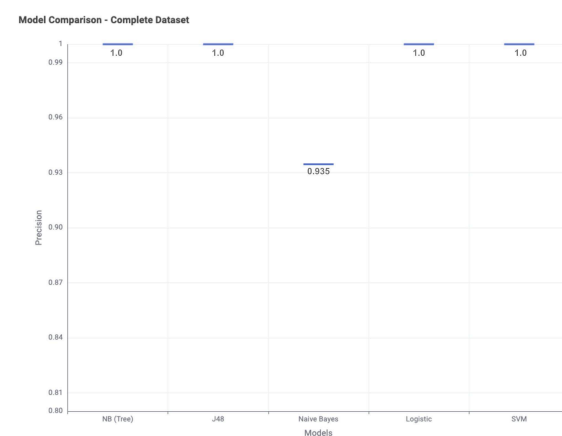


Figure 4 - Graph showing precision values obtained on the complete dataset with different supervised models.

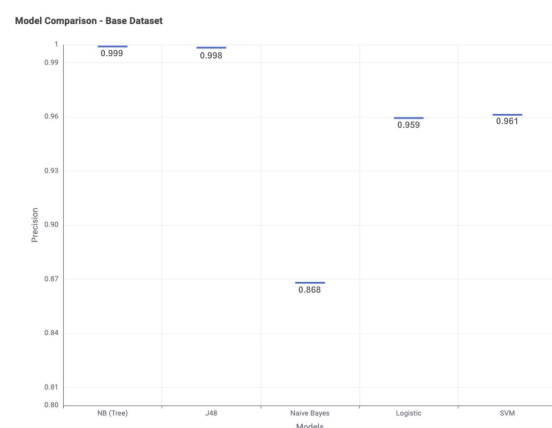


Figure 5 - Graph showing precision values obtained on the base dataset with different supervised models.

Base Dataset Containing Easy to Observe Variables

Reducing the dataset to solely the minimum set of easily observable variables results in a general decline in performance, albeit with varying degrees of intensity across models [Figure 5]. Decision tree-based classifiers maintain very high precision values, close to unity, indicating remarkable robustness to the reduction in information. Logistic Regression and SVM show a more moderate decrease, with precision around 0.96, whereas Naive Bayes exhibits a more marked drop, falling to approximately 0.87. These results suggest that the minimum set of variables retains a significant portion of the discriminative information but is insufficient, on its own, to guarantee perfect separation between edible and poisonous mushrooms for all models. Overall, the comparison between the complete and the minimum datasets allows us to delineate a performance range, representing the maximum and minimum levels of safety achievable by the classification models as a function of the amount of available information, according to our analytical process.

Incremental Phase

The incremental phase was applied to the two models that showed reduced performance when used on the base dataset but are capable of reaching a precision of 1.0 when applied to the full clean dataset: Logistic Regression and SVM [Figure 6]. The addition of the first two incremental variables, *stalk-surface-below-ring* and *stalk-surface-above-ring*, yields a performance improvement in both models, indicating that these features provide additional information relevant to distinguishing between edible and poisonous mushrooms. However, the observed increase is not monotonic for

both: Logistic Regression actually reports a decrease in precision of approximately 0.1 percentage points, suggesting the introduction of noise rather than useful information. A more pronounced increase in precision emerges with the introduction of the *ring-type* variable; this result suggests that this variable represents a feature capable of adding discriminative information for separating the two classes. Conversely, the addition of the *gill-spacing* variable does not produce a performance improvement; in the case of the SVM, a reduction in precision of 0.6 percentage points is even observed, indicating a likely addition of noise. The inclusion of the final variable, *odor*, proves to be decisive for both models, allowing them to reach a precision of 1.0 and thus completely align with the baseline obtained using the complete dataset. This result is consistent with the findings from the analysis of the association between variables and the target class [Figure 3], which identified odor as one of the features most strongly associated with mushroom edibility.

Turning now to the analysis of the clustering results [Figure 7], starting with the base dataset, the number of clusters required to achieve groups consisting solely of edible or poisonous mushrooms is 99. This number is significantly higher than the 23 clusters needed for the complete dataset; this result indicates that, given limited information, a clear separation between the two classes can only be obtained by fragmenting the dataset into a very large number of groups. This results in a loss of interpretability and practical utility for the clustering, especially considering that the dataset naturally consists of 23 mushroom species. Starting with the inclusion of the *stalk-surface-above-ring* variable, the number of groups required to obtain pure clusters settles at 23, a value that remains unchanged even with the subsequent addition of other variables. We

also validated the number of clusters using the Internal Index *Cophenetic Correlation Coefficient* which returns a value of 0.895 identifying a good fit of hierarchy to the data. This behavior suggests that, once a certain level of structural information is reached within the dataset, further feature enrichment does not yield additional simplification of the cluster structure in terms of purity. It is interesting to note that the introduction of *odor*, despite being decisive for achieving maximum precision in the supervised models, does not lead to a further reduction in the number of clusters necessary to obtain a purity of 1.

Conclusions

This analysis reinforces the notion that, for supervised models, increasing the complexity of the feature set does not automatically guarantee a performance improvement; rather, the informative contribution of individual variables is crucial. At the same time, the results show that it is possible to obtain performance comparable to the baseline established by the complete dataset by adding a limited number of highly discriminative variables. This provides useful insights for designing reliable models in contexts where feature observability is partial or costly. Clustering, on the other hand, reveals a complementary dynamic to that observed in supervised models: the progressive addition of variables allows for a marked initial reduction in the number of clusters, followed by a saturation phase. This suggests the existence of an information threshold beyond which the latent structure of the data, regarding the distinction between edible and poisonous mushrooms, is already sufficiently defined. A possible area for future work is to evaluate feature addition strategies from different perspectives that simulate various behaviors and real-world situations,

thereby diversifying the possible permutations among the features.

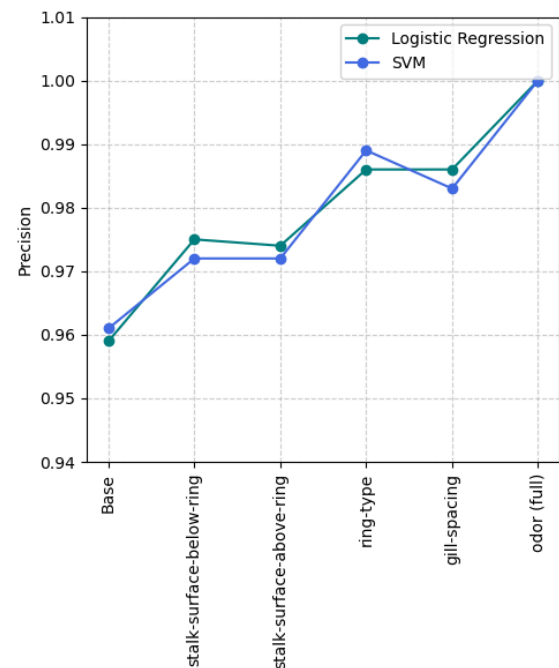


Figure 6 - Precision trend with respect to feature addition for Logistic Regression and SVM.

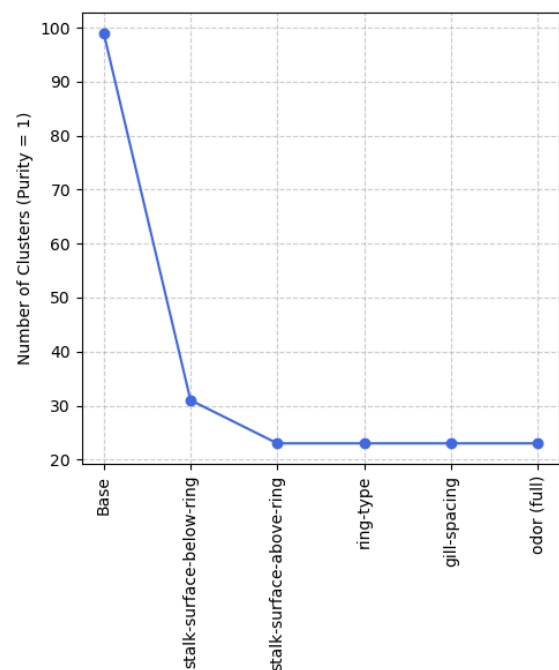


Figure 7 - Trend of the number of clusters with respect to feature addition at full purity (purity = 1).

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